

ActInf Textbook Group ~ Cohort 8 ~ Session 13

(Chapter 6 and overall) ~ 8/28/2025

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So, thank you again folks for joining. Uh, it's August 28th, 2025. This is cohort 8. We're starting the second half of the textbook reading group. And we know from the first chapter that uh the the structure of the book is such that the second half of the book is dedicated much more to applying active inference yourself to your own research as opposed to the first half which is more about setting up the kind of theoretical framework and and concepts and introducing you from things to what is a generative model, a generative process, what are action, perception What is the relationship of all of this to things like um theories of predictive processing to to the brazilian bayesian brain hypothesis? We had the low road and the high road to active inference. Um and and so it's here in the second half of the book that we're going to start learning how to actually apply this. It does mean that we will be getting uh much more in depth about u particular simulations uh and and models and and fitting models to to empirical data. Um it means like actually looking at simulations and and code and models that were already produced by the publication of this textbook and using them as sort of illustrative examples of some of the exemplar models that are used in active inference. uh not just to adhere to what's already been done, but to kind of provide us with a foundation on which we can uh build going forward uh for our own work. Um and so I've shared my screen. This is a bit of a throwback to a previous chapter, but it's just a fair reminder given that chapter 6 is a recipe for designing active inference models. um that whenever we kind of start out with considering like how should we build a model um what what is the work that's that's going to be entailed in any kind of project that applies these kinds of principles. Um first we recognize like we have what is the generative model or the agent. Those might be used kind of interchangeably throughout the textbook. And then on the other hand, we have the environment or the generative process. Um, and as we know from previous chapters, they're separated by a kind of uh marov blanket boundary. And so at at no point does the agent

necessarily have to have a model that is equivalent to the generative process. Um I I sometimes uh and forgive me but I'll sometimes use kind of silly analogies but um you know someone's saying knock on wood uh as an idiom that we use often to say like you know one does that for luck or by chance or something along those lines. It might be the case that knocking on wood, the action of that maybe consoles the person based on the beliefs they have about that idiom and and and how that maybe changes going forward how they view the situation because now they feel better that they knocked on wood or some kind of tick that they're they're trying to resolve for themselves or otherwise. Even though knocking on wood may not actually do anything to the generative process, actually it it might be in this case in the sense that it is impacting your own behavior going forward. Um, but there's no necessity that always knocking on wood actually aligns with some kind of real phenomena or generative process uh in reality outside of yourself. Right? So um that's kind of a crude example but it's just to say that whenever you set up a model there is no there's no necessity that it be entirely vertical to um the generative process in which it finds itself. Um and and anyone who engages in things like machine learning or deep learning or otherwise and works with very large models, blackbox models, etc. like they're they're more focused upon a certain kind of model architecture that allows for certain kinds of inference without it necessarily having any kind of onetoone interpretability with the environment. Right? So we have 7 billion uh hyperparameters uh for example you know some kind of very complex model that's very difficult to to parse out. Um and in any case so that this is just laying out a kind of recipe. The idea is well um first which system are we modeling? So later we'll get into examples of uh the kind of classic um mouse in a team uh example. So in that case it's very quick to kind of intuit it that the the agent here is the mouse and the environment is the teammates and everything else in which the the agent finds itself. Right? So that kind of aligns with the the image. You know the person's head here is now our mouse. the world is the T-mas and the the stimuli within and so on. Um and um secondly, what is the most appropriate form for the generative model? That's going to to require consideration over what kind of simulation are you building? Uh what is your agent doing? What is what is the context of what it is you're trying to build in uh chapter 7? So the in in that in that case the the mouse uh kind of begins at a starting point in a T-mase and it has three different things it can do. It can either uh move to the left arm of the T-mase or move to the right arm of the T-mase or it can move to uh the back of the T-mas that has an informative cue. Um, if it does that, then it gets an observation that kind of gives it a strong hint uh that the the stimuli it wants, which likened to a reward in reinforcement learning

or more literally we could say it's some kind of cheese or or some kind of sweet substance or otherwise it might be used in an actual laboratory. Uh so the the mouse has these three discrete options and it moves to three discrete locations. Uh it collects either a positive stimuli that being the cheese or a negative stimuli that being some kind of you know potentially negative shock or something that it doesn't like. Uh so so in that case we're looking at uh kind of a discrete state space, right? We're not looking at necessarily like what what what does the mouse see in its visual field and and sampling from the environment around it in a sense that it has to look through some large visual field and and pick out particular things and then some large continuous state space. It's much more discretized than that. It's also much more amendable to doing different kinds of uh and much easier to fit that behaviorally, right? Because if you were doing this in reality, you could track did the mouse move left, did it move right? Just much simpler than than having to say like what particular angle at what acceleration did it move in this precise direction. Um so that all of that is to say and that's a leadup into chapter seven which will be on discrete models. chapter eight which will be on continuous models that active inference of course can confront either one of those scenarios and forms of state spaces and ways of kind of uh chopping up or not time uh in terms of collecting like you know if you have a a discrete simulation that runs over discrete time steps versus a simulation that involves continuous streams of of data. you can already begin to imagine how these kinds of models could be applied to a broad variety of phenomena, not just you know mice and te- mases. Um in any case, so we think of what is the most appropriate form of the generative model. I already mentioned discrete continuous. Should there be a kind of hierarchical construction in which case maybe there's um a model at a lower level that more directly interfaces with the environment versus some higher level uh model that's maybe related to planning or you know decision making. Um you know kind of example would be you know the lower level could be the eyes that are visually sampling an environment. a higher level something related to reasoning. The prefrontal cortex, however it is that you find it best to kind of find a correlate for that could then be what decides uh you know a kind of discrete uh object it is seeing within the visual field as the data is sampled from the lower level. So that's the that's this kind of hierarchical message passing process could then work with both continuous and uh discrete data. Um I'm particularly fond of those. Um and so so once you decide what is the form of the model, what what what is the mouse in your in the case of of what you're setting up, then how do you after you set up the generative model, how do you set up the generative process? Well, um we already are considering with the teammates

like, well, we have a left arm, we have a right arm, we have an informative cue. So, you're just thinking of all of these different things that the environment itself kind of elicits um you know, it elicits uh a cue. If the agent moves to the space where there is a queue, it elicits positive stimuli in the left arm if that's indeed where it should be. So, this I really like this image just because of the kind of reciprocal aspect of of the the environment elicits observations and the agent makes predictions and carries out actions. Uh sometimes I phrase it as an action uh an agent's action is the environment's observation and the environment's action is the agent's observation. It's a really nice way to suddenly we have this isomorphic relationship between an agent environment. You could also think of that as an agent and another agent, right? Um and then you could just blow that up and have, you know, 30 or 100 agents within an environment. You have all of these sort of loops going on between them. Um so that's that's sort of the th those are kind of the key topics here. Um I there are already different things implied implied in what I said regarding like time scales of inference. Um this is related to some things in previous chapters but oftentimes in cases of learning um the one mo if you have a hierarchical model for example the lower level might learn faster than the than the higher model and so the higher order model kind of modulates uh the lower order model um different kinds of constructions like that. So well for sharing we'll uh just given the time uh we'll kind of open it up from there and uh cuz it's th this chapter is really just setting things up for anyone who already does computational modeling. None of this is going to be incredibly surprising to read uh for for anyone who who wants to set up a model. These are these are the kind of questions they're thinking about. It's just kind of putting it in the framework of active inference thinking like we're thinking generative processes and beliefs and states and so on. So, um, yeah. Did, um, so Benjamin, did you have any kind of questions? For example, not to put you on the spot, but >> no, no major questions. I this I remember this diagram from chapter 2, which is all I've read through so far. So, this is relevant to Yeah. what I was reading. It makes sense. No major questions. And I had just shared in the chat like a similar diagram that I reminded you of. >> Cool. Might check that out. See just world is system selfsynthesized by quantum networking. Yeah, >> just in terms of it being like a closed loop or something, but that's a much older paper. It's from 1980. >> 1980. Gotcha. I mean, it's >> Yeah. 88. >> 88. It it's it's so interesting that like like if you if you ever decide to take the time to kind of like skim around chapter 10, which arguably should be chapter 2 in my opinion, possibly. Uh but cuz chapter 10 is this sort of outro chapter, but really what it does is that it makes all these statements about the relationship between active inference and other fields.

just sort of in the spirit of making it clear that this is a a very interdisciplinary framework. Um I mean it it draws from principles of of cybernetics and control theory. Um we have aspects of of biology and reinforcement learning relationships with economics. >> And I only found this paper reading through like some Santa Fe Institute book of book of like good papers over the years and kind of chasing down a citation. So that that also exposed me to how interdisciplinary it can be. >> Nice. Yeah. Yeah. For sure. I mean all these things are implicated and so I mean anyone who tries to to to model vision in in active inference we are looking at things like the retina. We're looking at light. we're looking at um you know movement and and and so there are a lot of attempts to kind of if you if you want to model a phenomenon um people will set up for a generative process or environment like an entire you know set of kind of like physics in a in a simulation. So like um being able to simulate a a car accelerating up and down a a you know elevated plane that's at an angle such that there's friction and so on. Um I mean all of these it's just kind of that uh that nice philosophical sort of like everything in the in the universe is interconnected. Uh but and yet we're trying to zoom in on a piece of it uh and model it as we can. And I I think it's that um the concept of the mar of blanket which you'll find in I believe it's introduced roughly around chapter three or four but it's it's it starts to kind of pervade the first half of the book as you go forward. But um in some ways the idea of a markup blanket is kind of a useful tool for trying to say like well how do we delimit the agent or how do we delimit the the phenomenon of interest I'm looking at and the environment around. Um so yeah >> I'll peek ahead to chapter 10 and since you said it might be appropriate for chapter 2. Thanks for riffing on that. >> Yeah absolutely. I like pointing people to it because whenever you're getting into active inference, it's like you're coming from somewhere. Um, Daniel came here through uh an entomology studying uh ants and biology and I came here through through social science and an interest in kind of clinical uh psychotherapy and psychiatry. So it's just it it was useful for me to get to here through and then continued making me more interested um whenever I came here through more neuroscience and was looking at neuroimaging data and then I was like oh Fristen uh and company really get it um the way that they're trying to actually use findings from neuroscience to inform kind of bigger questions as well as giving a framework for doing computational modeling with that that Pier24 lacks a lot of kind of standard methods or frameworks and people just kind of have their own background and if they can't find a way of unifying those backgrounds at least to some degree it makes it much more difficult to have sort of like interdisciplinary dialogues. So yeah, um I think we'll it's completely okay if we wrap up uh a

bit early this time. We still let Ryan had to head out. Um, so >> I do actually have one more question based on what you were saying, which was like when you were talking about the maze, you were talking about a informative Q and I guess my question is like I think I understand what you're saying, but you mean the letter Q, right? Or do you mean a QE? >> Gotcha. No. Yeah, sorry. It it'd be c U. It's um >> ah, okay. Well, that makes sense as well. I thought you might might be there might be some jargon that is a thing called informative that is symbolized as cute but now I also understand what you mean by cu >> yeah no for sure it's it's a good um it's I mean while doing that kind of like a mouse in a team example I mean it's been you know it's been done many many times over in actual laboratories and empirical work um but it's not just used in this book to say like everyone needs to care about this kind of paradigm uh and and care about mice behavior. But uh here it's simply to it's a good example of it's an agent. The agent has multiple options available to it or things that it can do. Um and yet whenever we get into notions of planning and expected free energy which you'll start to come upon if not already in chapter 2 three >> yeah it it's it's there and this is the thing that was very interesting to me coming also from a slight like reinforcement learning agent-based modeling background was um um an agent who is driven to explore and find more information and minimize its uncertainty even though the action it's taking doesn't necessarily lead ex like expressly to a reward. Um, so I'm not sure on your familiarity with those kind any of those kinds of problems, but it's sort of like the explore exploit dilemma, which is a a very classic kind of debate in uh in social science for sure, but it's it's it's it's big in in reinforcement learning and machine learning generally because whenever a model gets stuck in a local optima um like you know it's found a suboptimal solution to a problem, but it doesn't know how to get out of that because it's learned to do, you know, um, in standard reinforcement learning, it's basically it's learned the best thing it can do given what it's been through thus far. So, why do anything else? Why doesn't why doesn't a human being just, you know, stay in a dark room as much as possible and minimize all of their uncertainty and, you know, eat food and and that's it. And, you know, they don't move on further beyond that. And um so expected free energy viewed just kind of mathematically as like an algorithm. It it has kind of an exploration term within it instead of it having to be a kind of hardcoded rule which is sometimes done in reinforcement learning. It'll say like you know the it's much more sophisticated nowadays but still sometimes people rely on like oh have the agent do something random 10% of the time. uh just just have them do something random. Uh just so that because >> entropy or entropy I guess in injection just to prevent >> Yeah. >> optimal stability or

something. >> Yeah, it's like a it's like a it's like an odd uh I would call it a band-aid, but it's uh it's not necessarily helpful in all cases. Um but uh so so yeah with with expected free energy it's like we you have within the algorithm a term that can differentiate relative to the model because everything is kind of relative to the model in active inference and modeling in general um you know it it it learns about its policies and so by eliciting new observations through trying new policies that it hasn't really tried before or maybe it hasn't tried them in particular contexts. Um it it it effectively tries it it estimates through inference what is the information gain that I get out of doing this. And um the the claim in the book which is slightly misspoken is that um information gain and and pragmatic value which you can think of as just going directly to a reward. Um it they get naturally balanced. They're not actually necessarily balanced in all cases, but the point is that both are computed simultaneously. And so you can score any any possible policy in active inference with with exploration and exploitation kind of combined. So there is no arbitrary rule here going on. There are no band-aids. Um it's just kind of kind of baked in. um could that get the agent theoretically into trouble? Maybe. Uh children make mistakes all the time, but uh is one funny way of phrasing that. But um yeah, so that's that's to me that's been one of the biggest sort of like oh like I I appreciate this entire notion and if it's if it could be traced back to any kind of empirical work that's been done um or with neuroimaging data or behavioral data or otherwise then you know I it's honestly one of the best ways of handling that explore exploit dilemma that I can possibly think of up to this point in time. So, um, that's sort of my take, but >> thank you. And you have to run to this other meeting, but I appreciate your help and added context and tips. So, hopefully see you at the next one. >> Yeah, absolutely. Thanks for thanks for stopping by. We'll see you next time. Bye. Well, take care.